Probability; Intro to Foundations of Inference

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MATH 240

Week 7, Lecture #3 — Ch. 3 (OI); 11

Independence

Two processes or events are said to be **independent** if knowing the outcome of one provides no useful information about the outcome of the other.

- Example: consecutive coin tosses
- Not an example: consecutive card draws from same deck

Marginal, Joint, and Conditional Probabilities

Suppose we want to understand the relationship between two events:

- M: movie earns a lot of money, and
- G: movie is G-rated

Then, we can calculate these probabilities:

1. **Marginal probability**: captures information about a single event/process at a time
2. **Joint probability**: considers how these two (or more) processes behave simultaneously
3. **Conditional probability**: encodes how the probability of one event changes given that we know that the second event has occurred

Marginal, Joint, and Conditional Probabilities

1. **Marginal probability**: captures information about a single event/process at a time
   - $P(M) = 0.180$
2. **Joint probability**: considers how these two (or more) processes behave simultaneously
   - $P(M \cap G) = P(M \text{ and } G) = 0.008$
3. **Conditional probability**: encodes how the probability of one event changes given that we know that the second event has occurred
   - $P(M|G) = P(M \text{ given } G) = \frac{P(M \cap G)}{P(G)} = 0.379$

Example: Movies

Rating	Century		Total
	20th	21st	
G	27	43	70
NC-17	11	2	13
Not Rated	10	21	31
PG	142	329	471
PG-13	192	917	1109
R	460	872	1332
Total	842	2184	3026

Table: Contingency Table of Movies based on Century of Movie Release and Rating

Multiplication Rule

We can multiply probabilities together depending on the situation.

1. If events A and B are not independent:
   - $P(A \cap B) = P(B|A)P(A)$
2. If events A and B are independent:
   - $P(A \cap B) = P(A)P(B)$

Example: Two Cards

Suppose the dealer reveal two cards to a player and the player is curious about the chance of having two aces.

1. Situation #1: Dealer chooses cards from same deck without replacement
   - Dependence occurs here: what the dealer shows after the first card is based on what the first card is

$$\begin{aligned} P(\text{2 aces}) &= P(\text{ace on second} \mid \text{ace on first}) P(\text{ace on first}) \\ &= \frac{3}{51} \times \frac{4}{52} \\ &\approx 0.0045 \end{aligned}$$

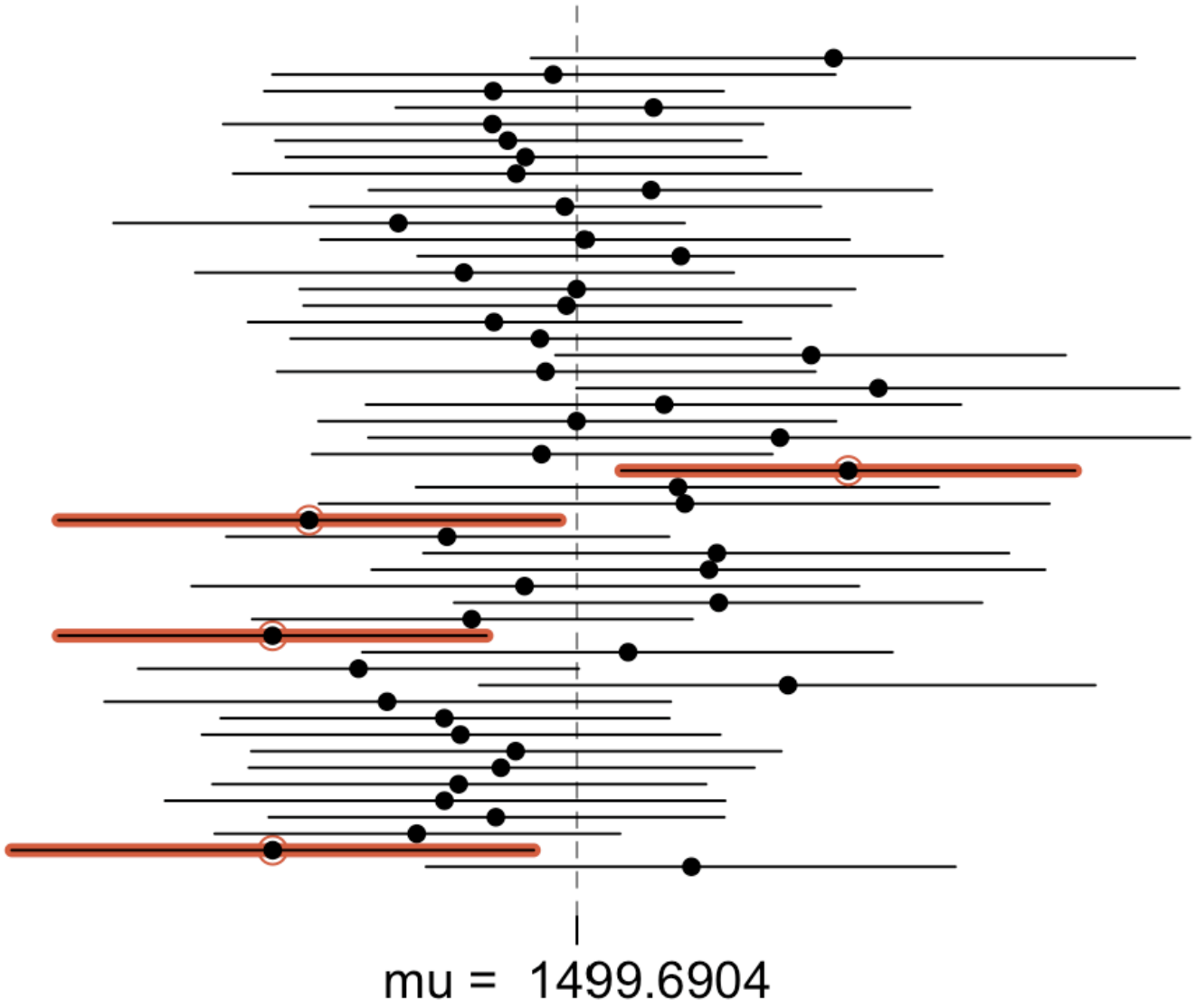
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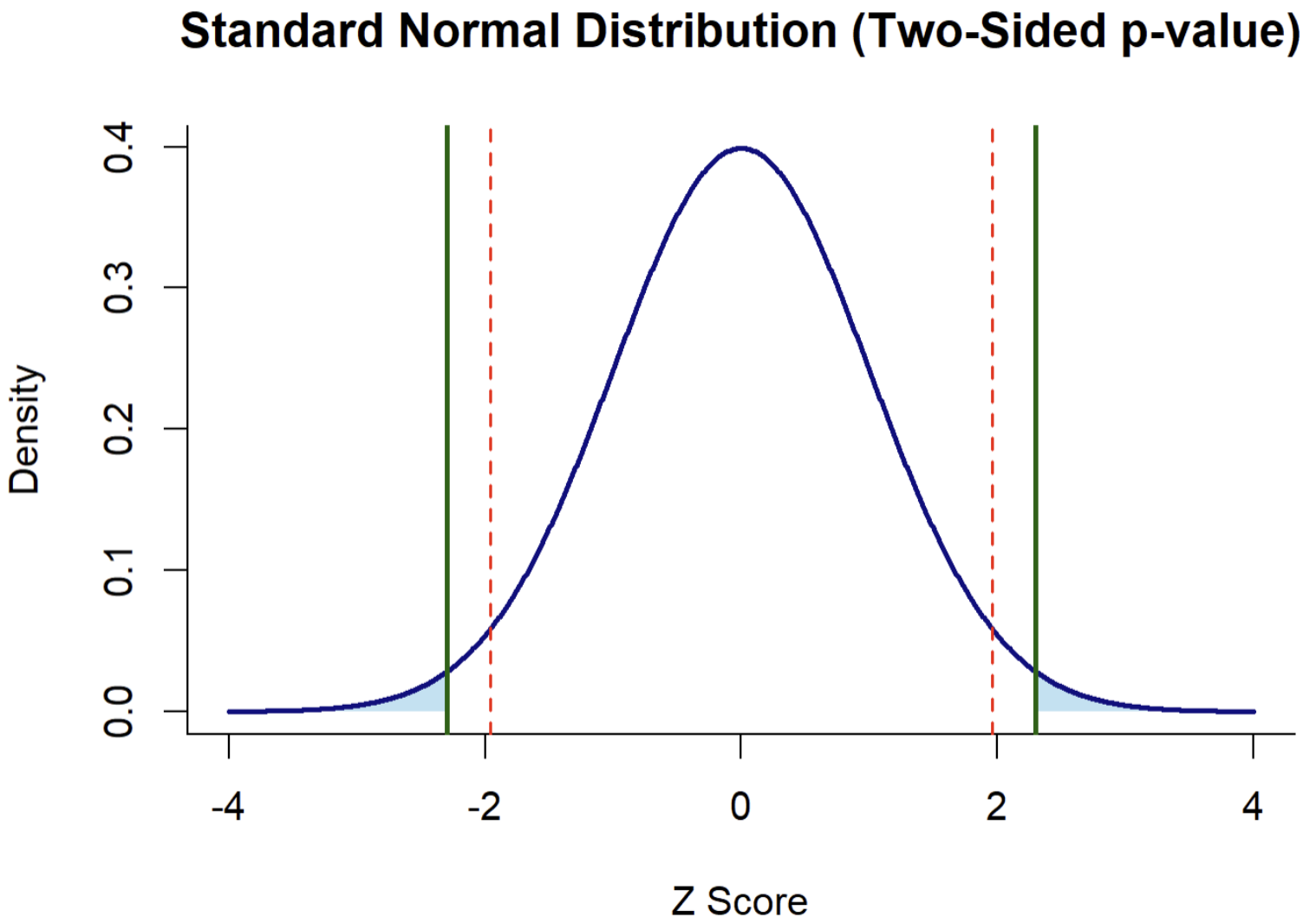
1. Situation #2: Dealer chooses card, reveals, then returns it to deck and reveals second card
   - Independent draws

$$\begin{aligned} P(\text{2 aces}) &= P(\text{ace on first})P(\text{ace on second}) \\ &= \frac{4}{52} \times \frac{4}{52} \\ &\approx 0.0059 \end{aligned}$$

Where Will Probability Appear Moving Forward?



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Third Unit: Inference

There is always an uncertainty with parameter estimates. As such, we need a system for understanding and quantifying this uncertainty. To do this, we will explore:

1. Hypothesis testing
2. Confidence intervals
3. Use of distributions (e.g. normal)
4. Decision errors (e.g. type I/II)

These will be the rest of the content leading into exam #2, which will be given on November 13th.

Motivation

We have some sample statistic that represents our best guess for the true population parameter. Additionally, we might have a sample statistic that deviates from our common belief.

- $\hat{p}$: an estimate of a proportion for population proportion p
- $\bar{x}$: an estimate of the population mean μ

However, we expect some variability for our estimates. This variability blurs our perception of whether these estimates are legitimate. Hence, we need a framework for testing our beliefs.

Motivation

The chance of rolling a 4 is $p = \frac{1}{6}$. The belief is that this chance holds up.

Suppose I suspect that the die I have is not fair. Then I would expect that the chance of rolling a 4 is not $p = \frac{1}{6}$.

However, I need to test out if this die is actually not fair. Consider rolling a die 24 times. I should expect to have 4 fours in these 24 rolls – but is it possible to get 5, 6, 7, or 8 fours?

Activity

In groups of 2-3: take a die and roll it 24 times. Record how many times you roll a four. Once that is done, take a post-it note and place it over the number of times you rolled a four on the chalkboard.

Hypothesis Testing

Hypothesis testing is the primary method for evaluating two competing possibilities, typically with relation of a sample statistic to some common belief for a population.

Hypothesis Testing

- Is there enough **evidence** from a sample to support there being a change in the population?
- Examples:
  - Drug testing
  - Components in a machine
  - Productivity along an assembly line
- Usually involves five steps
- Can involve population means, proportions, and variances

Steps of Hypothesis Testing

1. Define the problem
   - Parameter of interest
   - Null and alternative hypotheses
   - Level of significance
2. Collect data or run experiment
   - Determine sample statistic
3. Choose a model for testing
   - Appropriate test/model
   - Verifying conditions
4. Perform test/model
   - e.g. test statistic to get p-values and/or confidence intervals
   - e.g. randomization testing
5. Draw conclusions
   - Make decision on null/alternative hypotheses
   - Interpret decision in context of the problem